

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **K.Vignesh(192472154)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Vignesh

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.
Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment.
Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

Question 1

1. Scenario Understanding

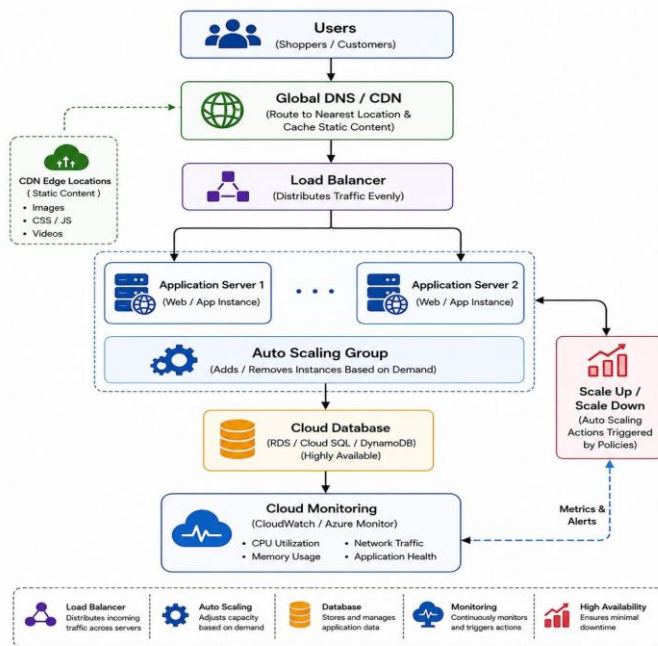
An e-commerce company experiences a sudden increase in website traffic during seasonal shopping events such as **Amazon Great Indian Festival**, **Flipkart Big Billion Days**, or **Black Friday**. Thousands or even millions of customers simultaneously browse products, add items to carts, and complete purchases. Without proper cloud infrastructure, servers can become overloaded, resulting in slow response times, failed transactions, and website downtime. Cloud computing addresses these challenges using **Auto Scaling**, **Load Balancing**, **Monitoring Services**, and **Content Delivery Networks (CDNs)**.

2. Challenges / Risks

The company faces several challenges during traffic spikes:

- Sudden increase in concurrent users.
- Server overload causing slow response times.
- Website downtime during peak sales.
- Failed payment and checkout transactions.
- Increased operational costs due to over-provisioning.
- Poor customer experience leading to revenue loss.

3. Cloud Architecture



4. Cloud Concepts Applied

Auto Scaling

Auto Scaling automatically adjusts the number of application servers according to user demand.

- During high traffic, new server instances are launched automatically.

- During low traffic, unnecessary instances are removed.

This ensures that adequate computing resources are always available while minimizing infrastructure costs

Load Balancing

A Load Balancer distributes incoming user requests evenly across multiple application servers.

Its functions include:

- Preventing server overload.
- Improving response time.
- Eliminating single points of failure.
- Ensuring continuous service availability.

As a result, all servers share the workload efficiently.

Monitoring Tools

Cloud monitoring services such as:

- Amazon CloudWatch
- Azure Monitor
- Google Cloud Monitoring

continuously monitor:

- CPU utilization
- Memory usage
- Network traffic
- Application response time
- Server health

When utilization exceeds predefined thresholds, monitoring tools automatically trigger Auto Scaling and notify administrators about system health.

5. Working of the Proposed Solution

1. Users access the e-commerce website.
2. The CDN delivers static content (images, CSS, JavaScript) from edge locations, reducing latency.
3. The Load Balancer distributes requests among available application servers.
4. Monitoring tools continuously observe CPU, memory, and network usage.
5. If traffic increases, Auto Scaling launches additional server instances.
6. When traffic decreases, extra instances are automatically terminated.
7. Customers continue shopping without interruptions or delays.

6. Real-World Example

During the **Amazon Great Indian Festival**, millions of customers simultaneously visit the website to purchase discounted products.

Amazon's cloud infrastructure uses:

- **Elastic Load Balancing (ELB)** to distribute incoming traffic.
- **Auto Scaling Groups** to launch additional EC2 instances during peak demand.
- **Amazon CloudWatch** to monitor resource utilization and trigger scaling actions.
- **Amazon CloudFront (CDN)** to deliver images and videos from the nearest edge location.

This architecture enables Amazon to process millions of user requests with minimal downtime and fast page loading.

7. Benefits

The proposed cloud solution provides:

- Efficient handling of unpredictable traffic spikes.
- High availability and fault tolerance.
- Reduced website downtime.
- Faster response time and improved user experience.
- Optimized cloud resource utilization.
- Lower operational costs through dynamic resource allocation.
- Improved business continuity during seasonal sales.

8. Summary Table

Cloud Component	Function
Auto Scaling	Automatically adds or removes servers based on demand
Load Balancer	Distributes incoming traffic evenly across servers
Cloud Monitoring	Monitors CPU, memory, traffic, and server health
CDN	Delivers static content from edge locations to reduce latency
Cloud Database	Stores product, customer, and order information

9. Conclusion / Justification

Cloud **Auto Scaling**, **Load Balancing**, **Monitoring Services**, and **CDNs** work together to create a highly scalable, reliable, and cost-efficient infrastructure for e-commerce applications. Auto Scaling dynamically adjusts computing resources according to traffic demand, while Load Balancing evenly distributes user requests to prevent server overload. Therefore, this cloud-based architecture minimizes downtime, enhances customer experience, optimizes operational costs, and ensures uninterrupted service during large-scale seasonal sales events.

Question 2

1. Application Scenario

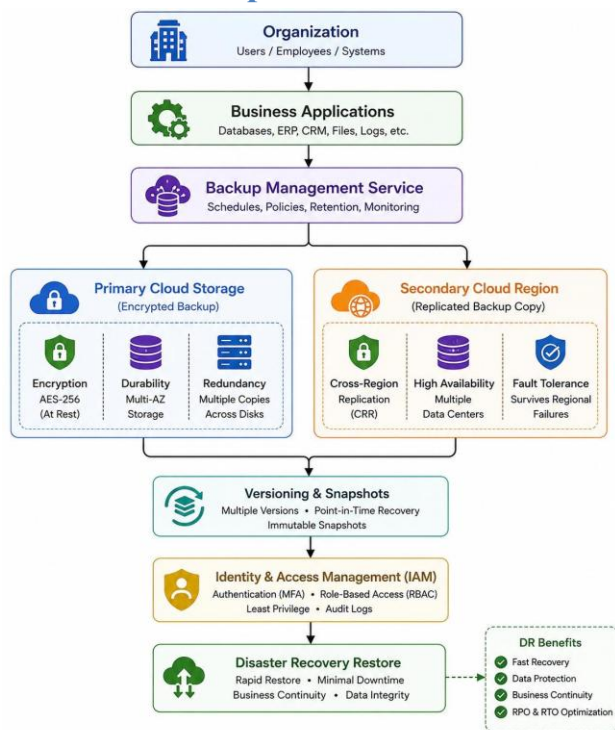
An organization generates large amounts of critical business data every day, including databases, application files, financial records, customer information, and system logs. To protect this data against accidental deletion, hardware failures, ransomware attacks, or natural disasters, the organization plans to use **cloud storage** for secure backup and disaster recovery. The cloud solution must ensure high durability, data redundancy, security, and cost efficiency.

2. Challenges / Risks

The organization faces several challenges:

- Loss of important business data due to hardware failure.
- Unauthorized access to backup files.
- High storage costs with traditional backup systems.
- Long recovery times during disasters.
- Risk of permanent data loss if backups are stored at a single location.

4. Cloud Backup Architecture



4. Cloud Concepts Applied

Cloud Storage Durability

Cloud providers store backup data across multiple storage devices and data centers. If one storage device fails, identical copies remain available in other locations.

This provides:

- High durability (typically 99.999999999% or "11 nines")
- Protection against hardware failures.
- Long-term data preservation.

Data Redundancy

Cloud storage automatically creates multiple copies of backup data.

Redundancy may include:

- Multiple disks within the same data center.
- Multiple Availability Zones.
- Cross-region replication.

Benefits include:

- Data availability during failures.
- Improved disaster recovery.
- Reduced risk of permanent data loss.

Encryption

Sensitive backup files are protected using encryption.

Encryption at Rest

- Data stored in cloud storage is encrypted using algorithms such as AES-256.

Encryption in Transit

- Data transferred between the organization and cloud storage is protected using TLS/SSL.

This prevents unauthorized users from reading backup data.

Access Control

Cloud providers use **Identity and Access Management (IAM)** to control access.

Security measures include:

- Multi-Factor Authentication (MFA).
- Role-Based Access Control (RBAC).
- Least Privilege Principle.
- Audit logging.

Only authorized administrators can access backup files.

5. Cost Advantages Compared to Traditional Storage

Cloud storage offers significant cost benefits:

- No need to purchase expensive physical storage hardware.
- Pay-as-you-use pricing model.
- Automatic scalability without purchasing additional equipment.
- Reduced maintenance and electricity costs.
- Lower backup administration expenses.
- Archive storage options for infrequently accessed data.

Compared to traditional storage systems, cloud storage reduces capital investment while improving scalability and reliability.

6. Disaster Recovery Example

Consider a hospital whose local servers are damaged due to flooding.

Since patient records are backed up in cloud storage:

- Replicated copies remain available.
- Backup versions can be restored immediately.
- Critical healthcare services resume quickly.
- Data loss is minimized.

This ensures business continuity and protects sensitive patient information.

7. Benefits

The proposed solution provides:

- High data durability.
- Automatic redundancy.
- Strong encryption and access control.
- Reduced storage costs.
- Faster disaster recovery.
- Improved business continuity.
- Secure long-term backup storage.

8. Summary Table

Cloud Component	Function
Cloud Storage	Stores backup data securely
Replication	Creates multiple copies across regions

Cloud Component	Function
Versioning	Maintains previous versions of files
Encryption	Protects data at rest and in transit
IAM	Controls user authentication and authorization
Disaster Recovery	Restores data during failures

9. Conclusion / Justification

Cloud storage provides a secure, scalable, and cost-effective solution for storing large volumes of backup data. By combining **high durability, automatic replication, encryption, access control, and disaster recovery mechanisms**, organizations can protect critical information from accidental deletion, cyberattacks, and hardware failures. Compared to traditional storage systems, cloud-based backup reduces infrastructure costs, improves recovery speed, and ensures continuous business operations, making it the preferred solution for modern organizations.

Question 3

1. Application Scenario

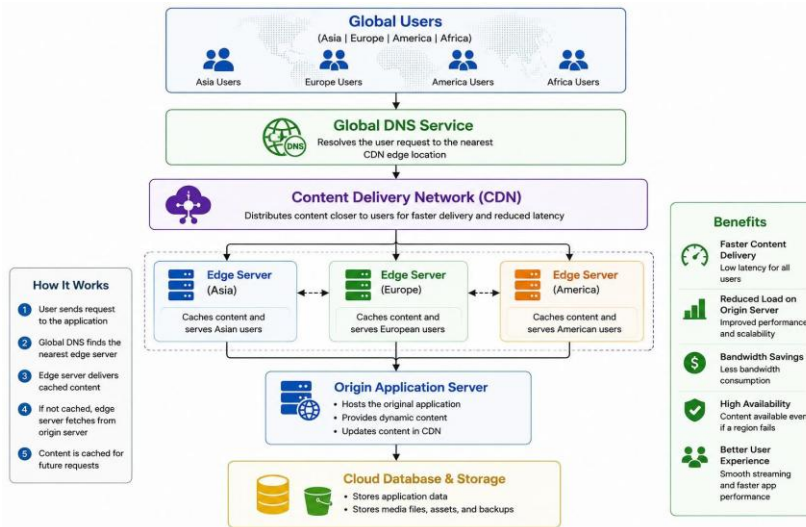
A mobile application company is launching its app worldwide. Users from different countries access images, videos, game assets, and APIs simultaneously. If all requests are served from a single server, users located far from the server experience **high latency**, slow loading times, and poor performance. To provide a fast and reliable experience, the company uses a **Content Delivery Network (CDN)** with globally distributed **Edge Servers**.

2. Challenges / Risks

The company faces several challenges:

- High latency for users located far from the main server.
- Slow loading of images, videos, and static resources.
- Increased bandwidth usage on the origin server.
- Poor user experience during peak traffic.
- Reduced application availability if one region experiences failures.

3. Global CDN Architecture



4. Cloud Concepts Applied

Content Delivery Network (CDN)

A CDN is a network of distributed servers that store cached copies of static content such as images, videos, CSS, JavaScript files, and game assets.

Instead of serving every request from the origin server, the CDN delivers content from the **nearest edge server**.

This results in:

- Faster content delivery.
- Lower latency.
- Reduced bandwidth consumption.
- Improved application performance.

Edge Servers

Edge servers are geographically distributed servers located closer to end users.

Their functions include:

- Delivering cached content locally.
- Reducing the distance data must travel.
- Decreasing response time.
- Reducing load on the origin server.

For example:

- Users in India receive data from the Asia edge server.
- Users in Germany receive data from the Europe edge server.
- Users in the USA receive data from the America edge server.

This significantly improves user experience.

Global Availability

Cloud providers such as **Amazon Web Services (AWS)**, **Microsoft Azure**, and **Google Cloud Platform (GCP)** maintain data centers across multiple geographic regions.

They support global availability through:

- Multiple cloud regions.
- Availability Zones.
- Global DNS.
- CDN services (CloudFront, Azure CDN, Cloud CDN).
- Load Balancing.
- Auto Scaling.

If one region becomes unavailable, traffic is automatically redirected to another healthy region.

5. Working of the Proposed Solution

1. A user opens the mobile application.
2. The request is routed through the Global DNS.
3. The CDN identifies the nearest edge server.
4. Cached content is delivered from the local edge server.
5. If the content is unavailable, the edge server retrieves it from the origin server and caches it for future requests.
6. Users receive content quickly with minimal delay.
7. During traffic spikes, the CDN distributes requests efficiently across edge locations.

6. Real-World Example

Netflix uses a global CDN to deliver movies and TV shows to millions of users.

When a user watches a movie:

- The video is streamed from the nearest CDN edge server.
- Frequently accessed content is cached locally.
- Auto Scaling and Load Balancers handle increasing traffic.
- Users experience smooth playback with minimal buffering.

Similarly, online games such as **PUBG Mobile** and **Call of Duty Mobile** use edge servers to reduce network latency and improve gameplay responsiveness.

7. Benefits

The proposed solution provides:

- Faster content delivery.
- Lower network latency.
- Reduced bandwidth consumption.
- Improved application performance.
- Better user experience worldwide.
- High availability.
- Fault tolerance.
- Efficient handling of global traffic.

8. Summary Table

Cloud Component	Function
CDN	Delivers cached content from nearby servers
Edge Server	Reduces latency by serving local users
Global DNS	Routes users to the nearest edge location
Origin Server	Stores the original application content
Cloud Storage	Stores media files and application assets
Load Balancer	Distributes incoming traffic efficiently

9. Conclusion / Justification

A **Content Delivery Network (CDN)** significantly improves application performance by delivering content from geographically distributed **edge servers**, reducing latency and minimizing bandwidth usage. Combined with **Global DNS, Auto Scaling, Load Balancing, and multiple cloud regions**, cloud providers ensure high availability and fast content delivery worldwide. This architecture is ideal for real-time streaming and gaming applications, where low latency and uninterrupted service are essential for providing an excellent user experience.

Question 4

1. Application Scenario

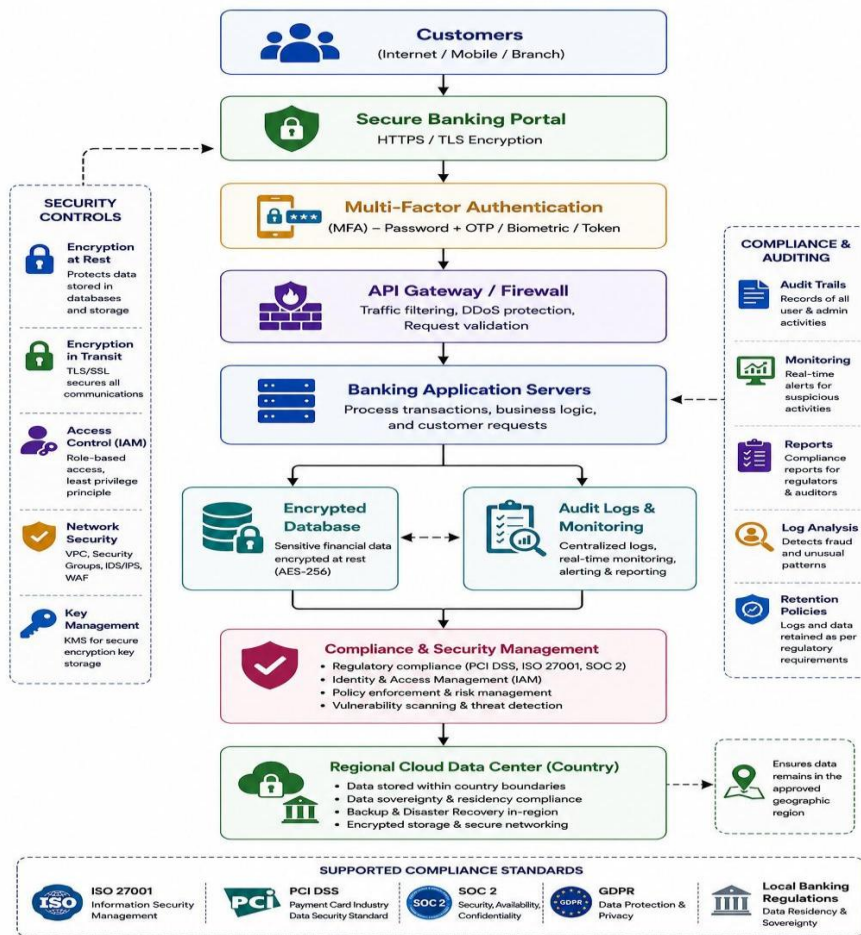
A financial institution handles highly sensitive information such as customer accounts, credit card details, online banking transactions, and financial records. Since this data is confidential and regulated by laws, the institution must use cloud services that ensure **security, compliance, auditing, and data protection**. The cloud infrastructure should also meet **data sovereignty** requirements by storing data in approved geographic locations.

2. Challenges / Risks

The institution faces several challenges:

- Unauthorized access to financial data.
- Data breaches and cyberattacks.
- Non-compliance with banking regulations.
- Lack of visibility into user activities.
- Data stored outside the country's legal jurisdiction.
- Financial fraud and transaction manipulation.

3. Secure Financial Cloud Architecture



4. Cloud Concepts Applied

Regulatory Compliance

Cloud providers support compliance by offering certifications and security standards such as:

- ISO 27001
- PCI DSS
- SOC 2
- GDPR (where applicable)

These standards help organizations meet legal and industry security requirements.

Auditing

Cloud auditing records every important activity performed within the cloud environment.

Examples include:

- User login history.
- File access.
- Configuration changes.
- Database modifications.
- Administrative actions.

Audit logs help detect suspicious activities and simplify compliance verification.

Logging and Monitoring

Cloud monitoring services continuously observe:

- User authentication attempts.
- Server performance.
- Network traffic.
- Failed login attempts.
- Security events.

Examples include:

- Amazon CloudWatch
- Azure Monitor
- Google Cloud Operations Suite

These tools generate alerts when unusual activities or security threats are detected.

Encryption

Cloud providers protect sensitive financial information using encryption.

Encryption at Rest

- Data stored in databases and storage services is encrypted using AES-256.

Encryption in Transit

- Communication between customers and cloud services is secured using TLS/SSL.

Encryption ensures that intercepted data cannot be read by unauthorized users.

Data Sovereignty

To satisfy legal requirements, financial institutions should:

- Store customer data only within approved geographic regions.
- Select cloud regions located in the required country.
- Restrict cross-border data transfers.
- Use regional backup and disaster recovery sites.

This ensures compliance with national financial regulations.

5. Working of the Proposed Solution

1. Customers securely access the banking application.
2. Multi-Factor Authentication verifies user identity.
3. API Gateway and Firewall inspect incoming requests.
4. Banking servers process transactions.
5. Financial data is encrypted before being stored.
6. Monitoring services continuously analyze logs and system activities.
7. Audit records are maintained for compliance verification.
8. Customer data remains stored within the country's cloud region.

6. Real-World Example

Consider an online banking application where a customer transfers money using internet banking.

During the transaction:

- User authentication is verified using MFA.
- Transaction details are encrypted using TLS.
- The database stores encrypted financial records.
- Every transaction is logged for auditing.
- Monitoring services detect suspicious login attempts or unusual transaction behavior.
- All customer data remains within the country's approved cloud region to satisfy banking regulations.

This ensures secure and compliant financial operations.

7. Benefits

The proposed solution provides:

- Strong protection against cyberattacks.
- Regulatory compliance.
- Secure financial transactions.
- Continuous monitoring and auditing.
- Protection of customer privacy.
- Legal compliance through data sovereignty.
- Improved trust and business continuity.

8. Summary Table

Cloud Component	Function
Multi-Factor Authentication	Verifies user identity securely
API Gateway & Firewall	Protects applications from unauthorized access
Encryption	Secures data at rest and in transit
Audit Logs	Records all user and system activities
Cloud Monitoring	Detects security threats and performance issues
Regional Cloud Data Center	Ensures data sovereignty and compliance

9. Conclusion / Justification

Cloud providers enable financial institutions to achieve **high security, regulatory compliance, and data protection** by combining encryption, auditing, monitoring, identity management, and regional data storage. Continuous logging and real-time monitoring help detect security threats quickly, while compliance certifications and data sovereignty measures ensure adherence to legal regulations. This architecture provides a secure, reliable, and compliant cloud environment for handling sensitive financial transactions.

Question 5

1. Application Scenario

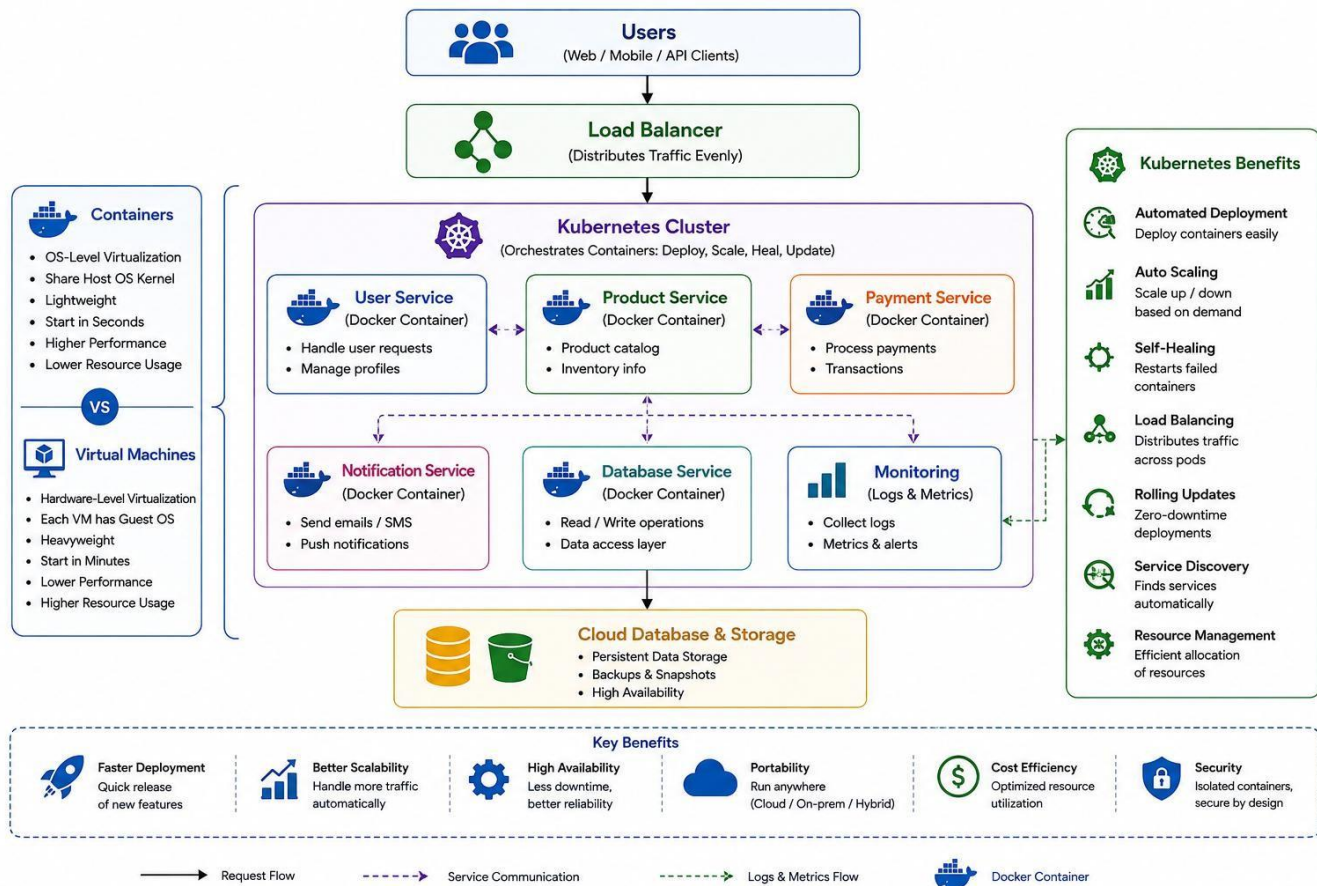
A software development team is building a cloud-native application using a **microservices architecture**. Instead of deploying the entire application as one large system, they divide it into independent services such as **User Service, Product Service, Payment Service, and Notification Service**. To deploy these services efficiently across different cloud platforms, the team adopts **Docker containers** and manages them using **Kubernetes**.

2. Challenges / Risks

The development team faces the following challenges:

- Consistent deployment across development, testing, and production environments.
- Efficient utilization of computing resources.
- Managing multiple application services.
- Scaling applications during increased user demand.
- Migrating applications across different cloud providers without modification.

3. Containerized Application Architecture



4. Cloud Concepts Applied

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Virtualization	OS-Level	Hardware-Level
Operating System	Shares host OS	Separate OS for each VM
Startup Time	Seconds	Minutes
Resource Usage	Low	High
Performance	Faster	Slightly slower
Portability	High	Moderate

Containers are lightweight because they share the host operating system, whereas virtual machines require a separate guest operating system, increasing resource consumption.

Kubernetes Orchestration

Kubernetes automates container management by providing:

- Automated deployment.
- Auto Scaling.
- Load Balancing.
- Self-healing (automatic restart of failed containers).
- Rolling updates.
- Service discovery.
- Resource scheduling.

These features simplify application deployment and management in cloud environments.

Portability Across Cloud Providers

Containers package the application along with its dependencies, allowing the same container image to run on different cloud platforms without modification.

Examples:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

This reduces vendor lock-in and simplifies cloud migration.

5. Working of the Proposed Solution

1. Developers package each microservice into a Docker container.
2. Container images are stored in a container registry.
3. Kubernetes deploys the containers across multiple nodes.
4. A Load Balancer distributes user requests to the appropriate service.
5. Kubernetes continuously monitors container health.
6. If demand increases, Kubernetes automatically launches additional container instances.
7. Failed containers are automatically restarted without affecting the entire application.

6. Real-World Example

An **online food delivery application** such as **Swiggy** or **Zomato** uses a microservices architecture.

Different services include:

- User Service
- Restaurant Service
- Order Service
- Payment Service
- Delivery Tracking Service
- Notification Service

Each service runs inside its own Docker container and is managed by Kubernetes. During peak meal hours, Kubernetes automatically scales services such as **Order** and **Payment** to handle increased customer traffic while maintaining smooth application performance.

7. Benefits

The proposed solution provides:

- Faster application deployment.
- Better resource utilization.
- Improved scalability.
- High availability.
- Easy portability across cloud platforms.
- Simplified application management.
- Reduced infrastructure costs.
- Faster software updates and releases.

8. Summary Table

Cloud Component	Function
Docker Containers	Package applications with dependencies
Kubernetes	Orchestrates deployment, scaling, and management
Load Balancer	Distributes traffic among services
Container Registry	Stores Docker images
Cloud Database	Stores application data
Monitoring Service	Tracks container health and performance

9. Conclusion / Justification

Containerization provides a lightweight and efficient approach to deploying cloud-native applications by reducing resource consumption and improving startup performance compared to virtual machines. Kubernetes simplifies deployment through automated orchestration, scaling, load balancing, and self-healing capabilities. The portability of containers enables applications to run consistently across different cloud providers, reducing vendor lock-in and simplifying migration. Therefore, containerization combined with Kubernetes is an ideal solution for modern microservices-based cloud applications requiring scalability, flexibility, and high availability.